

Multiple Choice (10 marks)

1. Let `l = [1,2,3,4]`. What is `sum(l)` in Python 3?
 - (a) 4
 - (b) 1
 - (c) 10
 - (d) 8
2. In Python 3, let `r = lambda q: q * 2`. What is `r(3)`?
 - (a) 2
 - (b) 6
 - (c) 3
 - (d) 1
3. In Python 3, what is `['a', 'Chemistry', 0, 1][1]`?
 - (a) a
 - (b) Chemistry
 - (c) 0
 - (d) 1
4. Which types of functions grow the slowest?
 - (a) $O(N^{1/2})$
 - (b) $O(N^{1/4})$
 - (c) $O(N^{1/N})$
 - (d) $O(N)$
5. Which of the following is correct about Python?
 - (a) It supports automatic garbage collection.
 - (b) It can be easily integrated with C, C++, COM, ActiveX, CORBA, and Java.
 - (c) Both of the above.
 - (d) None of the above.

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, converting a list to a set retains all elements, including duplicates, resulting in {1, 2, 2, 3, 4}.
7. True or False: In Python 3, the function `int(x [,base])` is used to convert a string to an integer.
8. True or False: The minimum value in the list `l = [1, 2, 3, 4]` is 1 in Python 3.
9. True or False: In Python 3, the expression `['a', 'Chemistry', 0, 1][-3]` evaluates to 'Chemistry'.
10. True or False: In Python 3, accessing the first element of the tuple `( 'abcd', 786 , 2.23, 'john', 70.2 )` returns 'abcd'.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the characteristics of phishing attacks and explain why an email from your bank asking you to verify a transaction by calling a number is least indicative of such an attack.
12. Discuss how heuristic methods can enhance the efficiency of programs, specifically in finding the shortest driving route between two locations on a map.

End of Paper